

Classification of the 100 Benchmark Patches by Solver-RMSE Ratio

1 Criterion and counts

For each patch P with N_P pixels, define the two whole-patch root mean square errors

$$d_{\text{SOLVER(GT)}}(P) = \sqrt{\frac{1}{N_P} \sum_{p=1}^{N_P} \left(\hat{R}_{P,\text{SOLVER(GT)}}(p) - R_P(p) \right)^2}, \quad (1)$$

$$d_{\text{CONVEX SOLVER}}(P) = \sqrt{\frac{1}{N_P} \sum_{p=1}^{N_P} \left(\hat{R}_{P,\text{CONVEX SOLVER}}(p) - R_P(p) \right)^2}. \quad (2)$$

The classification ratio is

$$q(P) = \frac{d_{\text{CONVEX SOLVER}}(P)}{d_{\text{SOLVER(GT)}}(P)}. \quad (3)$$

A patch belongs to **Case 1** when $q(P) \geq 2$: the Convex Solver's RMSE is at least twice the RMSE of SOLVER(GT), so SOLVER(GT) is much closer to the ground truth. A patch belongs to **Case 2** when $q(P) < 2$: the two RMSE values are comparatively close. The equality is assigned to Case 1 by definition.

Applying this rule to all 100 patches gives

Classification	Number of patches	Percentage
Case 1 ($q \geq 2$)	24	24.0%
Case 2 ($q < 2$)	76	76.0%
Total	100	100.0%

The observed ratios range from 0.9846 to 11.7670. There is no patch in the interval between the largest Case 2 ratio (1.9767) and the smallest Case 1 ratio (2.6310).

2 Distribution of the ratio

Table 1 reports the distribution of $q(P)$ across all 100 patches, independently of the assigned case. The intervals have width one and are left-closed and right-open.

Ratio interval	Number of patches	Percentage
[0, 1)	11	11.0%
[1, 2)	65	65.0%
[2, 3)	1	1.0%
[3, 4)	4	4.0%
[4, 5)	5	5.0%
[5, 6)	2	2.0%
[6, 7)	4	4.0%
[7, 8)	1	1.0%
[8, 9)	1	1.0%
[9, 10)	1	1.0%
[10, 11)	2	2.0%
[11, 12)	3	3.0%
Total	100	100.0%

Table 1: Distribution of $d_{\text{CONVEX SOLVER}}/d_{\text{SOLVER(GT)}}$ in bins of width one.

3 Patch assignments

The common identifier prefix `RAD_OPERA_HOURLY_RAINFALL_ACCUMULATION_` is omitted below.

3.1 Case 1: 24 patches

202301190300_patch001	202301241700_patch000
202301210300_patch000	202301241700_patch002
202301210800_patch002	202301242200_patch001
202301210900_patch000	202301250200_patch001
202301211800_patch000	202301260000_patch000
202301211900_patch000	202301270400_patch000
202301211900_patch001	202301270700_patch000
202301222100_patch000	202301272100_patch001
202301230100_patch000	202301280900_patch000
202301231300_patch001	202301281000_patch001
202301240500_patch000	202301282200_patch000
202301241500_patch000	202301282200_patch001

3.2 Case 2: 76 patches

202301180600_patch000	202301212300_patch000	202301250300_patch000
202301180900_patch002	202301220000_patch001	202301250500_patch000
202301181200_patch003	202301220100_patch001	202301251000_patch001
202301181300_patch002	202301221200_patch000	202301251300_patch000
202301181900_patch000	202301230600_patch003	202301251600_patch002
202301190000_patch000	202301231200_patch001	202301252100_patch000
202301190300_patch000	202301231200_patch002	202301252200_patch000
202301190700_patch000	202301231200_patch003	202301260200_patch000
202301190800_patch001	202301231400_patch002	202301260200_patch001
202301190800_patch002	202301231400_patch003	202301260600_patch000
202301190900_patch001	202301231600_patch001	202301260800_patch001
202301192000_patch000	202301231700_patch000	202301260900_patch000
202301200400_patch000	202301231900_patch000	202301260900_patch001
202301200800_patch001	202301240700_patch000	202301270800_patch000
202301201300_patch000	202301241500_patch001	202301270900_patch000
202301201600_patch002	202301241900_patch001	202301271000_patch001
202301211500_patch000	202301250200_patch000	202301271800_patch000

202301272300_patch000
202301272300_patch002
202301280000_patch001
202301280700_patch000
202301280800_patch000
202301281900_patch000
202301282100_patch000
202301290800_patch000
202301291100_patch000

202301291700_patch000
202301291900_patch000
202301292200_patch002
202301292300_patch001
202301292300_patch003
202301300000_patch001
202301300200_patch001
202301301000_patch000
202301302200_patch000

202301302200_patch001
202301302200_patch002
202301310000_patch001
202301310200_patch000
202301310900_patch000
202301311600_patch000
202301312000_patch001

4 Boundary heatmaps

The following figures show the two patches closest to the threshold from either side. All rainfall panels within a figure use the same color scale. Relative errors are computed as $(\hat{R} - R)/R$ for rainy ground-truth pixels with $R \geq 0.6$ mm/h; non-rainy pixels are gray. To keep the spatial error patterns legible, the symmetric error scale is clipped at the shared 99th percentile of the absolute relative errors from the two solvers.

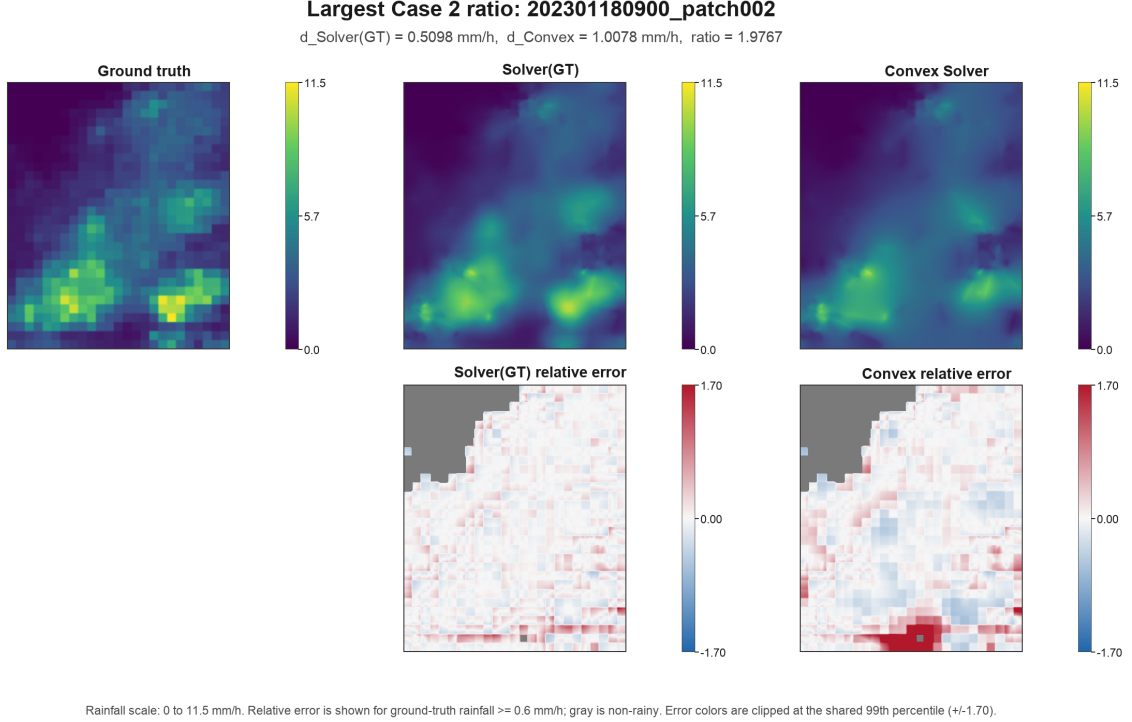
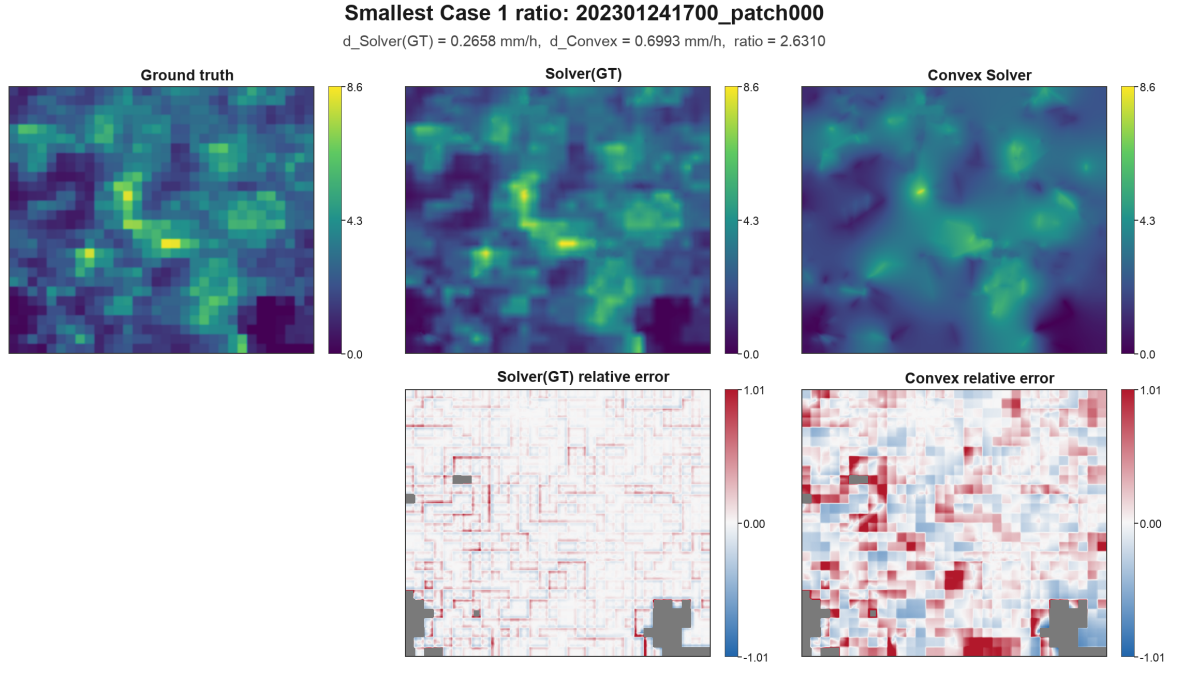


Figure 1: Largest ratio in Case 2: 202301180900_patch002. Here $d_{\text{SOLVER(GT)}} = 0.5098$ mm/h, $d_{\text{CONVEX SOLVER}} = 1.0078$ mm/h, and $q = 1.9767 < 2$.



Rainfall scale: 0 to 8.6 mm/h. Relative error is shown for ground-truth rainfall $\geq 0.6 \text{ mm/h}$; gray is non-rainy. Error colors are clipped at the shared 99th percentile (± 1.01).

Figure 2: Smallest ratio in Case 1: 202301241700_patch000. Here $d_{\text{SOLVER(GT)}} = 0.2658 \text{ mm/h}$, $d_{\text{CONVEX SOLVER}} = 0.6993 \text{ mm/h}$, and $q = 2.6310 \geq 2$.